Classification, Decision Trees & Regression Summary with Answers

1. Classification Basics
Goal: Learn a model that maps input features (x) → output categories (y).
Training vs Test: Training fits the model; Test evaluates it.
Independent vs Dependent: X are predictors; y is target.
Examples: Tumor type (benign/malignant), loan default (Yes/No).
Difference from Regression: Classification = discrete; Regression = continuous.

2. Decision Tree Structure
Decision node → test attribute; Leaf node → final class.
Non-parametric: no predefined equation.
Overfitting → tree too complex; Pruning reduces it.
Different trees can classify same data correctly.

3. Impurity Measures
Entropy, Gini, and Error measure homogeneity.
Measure	Formula	Result Example
Entropy	$-\Sigma\, p \cdot \log\blacksquare(p)$	1.571
Gini	$1 - \Sigma\, p^2$	0.66
Error	$1 - \max(p)$	0.60
Lower value = more pure node.

4. Splitting and Attribute Types
- Binary split: 2 groups (Income > 80K?).
- Multi-way: all unique values.
Nominal → unordered, Ordinal → keep order, Continuous → threshold split.
Discretization = converting continuous into ordered ranges.

5. Tree Induction Algorithms
Algorithm	Impurity	Note
ID3	Entropy	early model
C4.5	Entropy + Gain Ratio	improved ID3
CART	Gini	sklearn default
Greedy = best local split only, no backtracking.
Advantages: easy to interpret, handle mixed data, non-linear relations.

6. Ensemble Methods
Bagging vs Boosting
Type	Idea	Example
Bagging	random subsets averaged	Random Forest
Boosting	sequential, fix previous errors	AdaBoost, XGBoost
Random Forest → less overfitting.

7. Linear Regression + GridSearchCV
Regression = continuous prediction.
Scaling normalizes features.
GridSearchCV = find best parameters with cross-validation.
Cross-validation prevents overfitting.

8. Pandas Essentials
```python
import pandas as pd
df = pd.read_csv('data.csv')
df = df.dropna()
df.select_dtypes(include='number')
df[df['Income'] > 80000]
df['Income'].fillna(df['Income'].mean(), inplace=True)
```
.loc[] → label-based, .iloc[] → index-based.

9. Real-world Mini Challenge
Age	Salary	OwnsCar	BuysInsurance
25	30K	No	No
40	80K	Yes	Yes
35	60K	No	Yes
28	40K	No	No
50	90K	Yes	Yes

Entropy(root) = 0.971 (Yes=3, No=2).
Best split = OwnsCar.
New case (38y, 70K, NoCar) → BuysInsurance = Yes.